Forum: **ECOSOC**

Issue: **The question of governing transboundary water resources and preventing hydro-political conflict in arid regions facing climate change.**

Student Officer: **Caia Wang, Hiroyasu Kato**

Position: **Deputy Chair**

Introduction

Water is not only a necessity of human life, but how it's being utilised also determines the stability of a region and society, especially in dry regions. Currently, water scarcity is intensifying, with World Bank sources estimating that 2.2 billion people worldwide lack access to safely managed drinking water, and annual freshwater losses amount to a staggering 324 billion cubic metres. The World Research Institute also projects that water use in Sub-Saharan Africa will increase by 163% by 2050, with 25 nations predominantly in the Middle East and North Africa, including countries such as Bahrain, Cyprus, Kuwait, Lebanon, Oman, and Qatar, facing levels of acute water stress. According to United Nations and World Bank sources, approximately 700 million people are at risk of displacement due to droughts. Additionally, the poor harvests that follow dry seasons and food shortages further exacerbate the issue, with notable examples including 2017, when over 20 million people from North Africa and the Middle East were forced to migrate and seek refuge due to the accompanying conflicts that broke out from the scarcity of resources.

With global freshwater resources becoming increasingly scarce due to climate change, the dilemma pushes nations closer to the brink of tolerance and endurance, and water wars are not merely imminent but are already occurring. To acquire and secure control over such precious resources, nations are willing to sever diplomatic relations with each other and perhaps even enter a state of war. The organisation Circle of Blue reports a spike in water conflicts, from 231 incidents in 2022 to 347 incidents in 2023. This shows the extent and degree in which nations can become desperate over controlling aqueous commodities, and demonstrates the necessity of transboundary governance over water resources; considering that over 40% of the world’s population lives along transboundary expanses, and only a fifth of the countries included have bilateral or multilateral agreements over the usage of such resources, as shown in a UN World Water Development Report published by UNESCO in 2024.

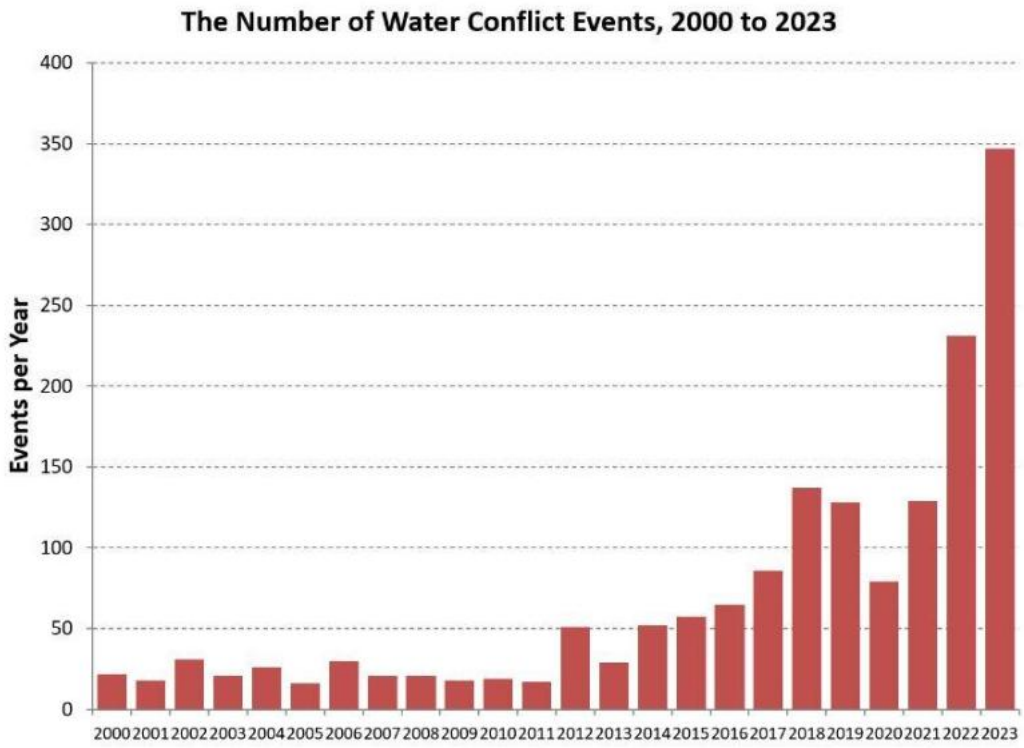

***Figure 1**, Bar chart of the number of water conflict events, 2000 to 2023, from Circle of Blue*

Resolving hydro-political conflicts cannot be solely relied upon through international action; regional frameworks designated by riparian countries have proven effective, from the mid-20th century to the present day. To prevent escalation of tensions, upstream and downstream nations maintain constant channels of dialogue and communication to facilitate understanding and negotiate aspects regarding the use of water, and even if diplomatic corridors are severed temporarily, members of the framework are usually bound to the agreement and would have to enforce the guidelines stated. Examples, such as the Indus Water Treaty and other systems covered in this report, demonstrate how the governance of transboundary waters, backed by documents, agreements, frameworks, and treaties, fosters solidarity in conserving and developing water resources sustainably.

Definition of Key Terms

Transboundary Water Resources

Bodies of water, including rivers, lakes and aquifers that are bordered and shared by more than one country.

Hydro-Political Conflict

Political disputes between countries regarding the status of shared water resources.

Water Security

The capacity of a population to safeguard sustainable access to adequate quantities of acceptable quality water for sustaining livelihoods, human well-being, and socio-economic development, for ensuring protection against water-

borne pollution and water-related disasters, and for preserving ecosystems in a climate of peace and political stability. (United Nations)

Integrated Water Resources Management (IWRM)

Promoting the coordinated development and management of water, land and related resources to maximise economic and social welfare equitably, without compromising the sustainability of vital ecosystems. (United Nations Environment Programme)

Water Stress

The inability to provide water or access to it to the point where human needs cannot be met.

Watercourse

The Convention on the Law of the Non-Navigational Uses of International Watercourses defines the term "watercourse" as "A system of surface waters and groundwaters constituting by virtue of their physical relationship a unitary whole and normally flowing into a common terminus." Essentially, this means that all groundwater and surface water that are hydrologically connected are to be treated as a single system and body of water.

Riparian

A state, community, or party that lies along and shares a river, lake, or other watercourse, possessing legal or geographical adjacency to that water body. (Convention on the Law of the Non-Navigational Uses of International Watercourses). The denotation refers to states with legitimate political rights and responsibilities regarding a water body.

Non-Consumptive Use:

A way in which water from a hydrological system is extracted so that its resources are not permanently diminished in quantity. Examples include dams, fishing, navigation, and transport.

Flow time

The control of when water is released downstream, typically in hydroelectric infrastructures such as dams and reservoirs. This leads to upstream riparians gaining an advantage in influencing irrigation cycles and downstream reservoir refill periods, which may result in political disagreements among other riparians.

Headwater

The region in which a river system originates. This can include uppermost streams, snowmelt zones, and glaciers, where freshwater flows down.

Tributaries

Streams and smaller rivers that feed into the main river.

Equitable Utilisation:

A principle that promotes shared natural resources being used fairly and reasonably, whilst also respecting other parties that depend on such resources.

Shortfall

The gap between available water resources to be utilised and extracted, and water demand. A shortfall typically refers to an insufficient supply of water to sustain a region.

Background

Humanity has been faced with issues regarding the supply and demand of water for millennia. In recent years, the governance and ownership of transboundary waters have deteriorated, becoming a significant problem, especially in regions with scarce water resources. As climate change has become more serious, water security has become increasingly important, and the situation has become more severe. Furthermore, development has accelerated in water-scarce areas, resulting in significant population growth and increased water stress. Since 1960, water withdrawals from freshwater sources have steadily increased, placing stress on the limited freshwater resources. In 2023, the UN's World Water Development Report revealed that 26% of the population, or 2 billion people, still lack access to clean drinking water. Drinking water is only one part of the issue; governments use water resources for agriculture, electricity and manufacturing. All these uses of water make it an invaluable resource, which is a catalyst for conflict and something that must be addressed if this entire issue is to be resolved.

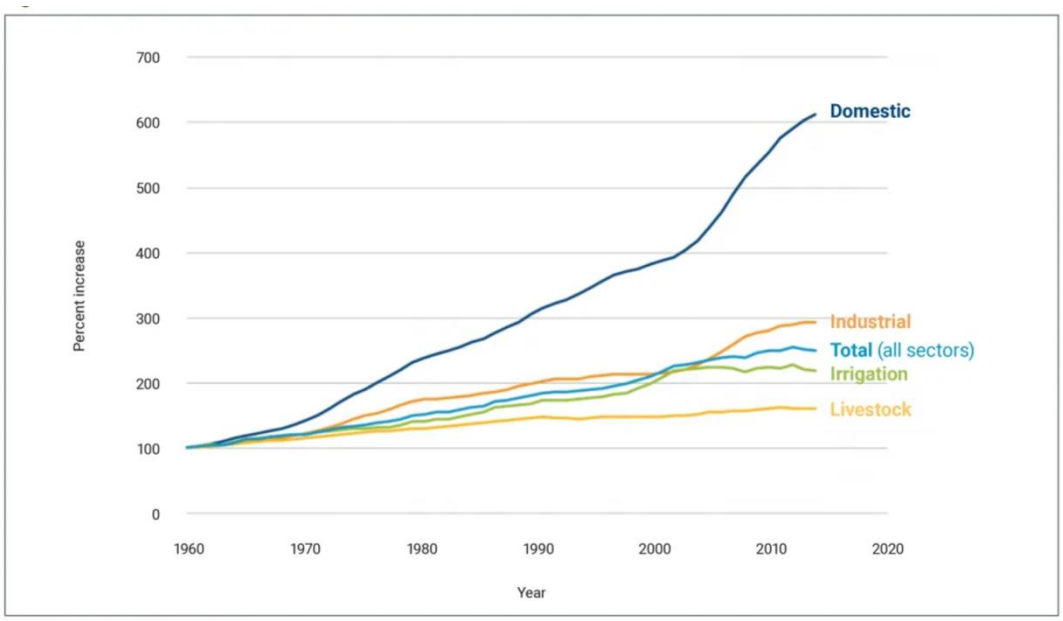

Figure 2: The increase of water withdrawals for different sectors, 1960 to 2020 (Otto and Schleifer, 2020)

Even non-consumptive use can pose a serious threat to many nations, as the construction of dams or transportation can significantly redirect or alter the amount of water. Upstream nations, known as upstream riparians, can exert their control over the flow time, allowing them to have influence and power, which may upset downstream riparians. Many nations are riparian states and face challenges when governing transboundary water resources. Political tensions can rise rapidly in the event of a dispute, and these problems are especially prevalent when there are no pre-existing frameworks to govern nations.

The issue of water scarcity and its impact on transboundary water resources is prevalent in many member nations, with notable examples including Egypt, Cambodia, India, and Pakistan. It is difficult, if not impossible, to resolve this issue without new frameworks in place, as climate change and extreme weather will only make these

water resources more valuable. In arid areas with large populations, there is additional pressure on governments to devise a solution, even if it negatively impacts other riparians. Rivers are especially contested; they can serve as a pathway for trade and transport, but also remain an important freshwater source. Although they are generally quite large when multiple nations access them, the direction and amount of water in the river can alter from season to season, leading to challenges in the nations that depend on them. Nations must find a way to utilise these water sources efficiently and, when possible, find alternatives to common practices that use less water. Examples of this can be crops that require less water or sanitation facilities that minimise water use. This will enable water to be redirected to areas where it is necessary, thereby reducing conflict and fostering multilateral cooperation.

The Grand Ethiopian Renaissance Dam (GERD)

During September of 2025, Ethiopia celebrated the official operation of the Grand Ethiopian Renaissance Dam (GERD), a massive hydropower project on top of the Blue Nile, the largest tributary that originates in Ethiopia and joins with the White Nile in Sudan, contributing around 85% of total water resources to the remaining downstream, which flows towards Egypt. The GERD is estimated to provide around 5,510 MW of hydroelectric power to Ethiopia, enabling it to supply electricity to 120 million people, making it the largest dam on the Blue Nile. However, over the 14 years of the GERD's construction, downstream nations such as Sudan and Egypt have expressed strong opposition to the project.

The project has become a source of regional tension in the Horn of Africa and North-East Africa. Sudan and Egypt have released joint statements claiming that the GERD has “breached international law and would cause grave consequences to the two downstream countries,” and calling for the establishment of multilateral treaties between the three states regarding the shared management of the Blue Nile. Meanwhile, Ethiopia claims that the GERD for Ethiopians is a “celebration of a generational aspiration” and insists it is a “regional dam with collective aspirations”, Over the years, Egypt and Sudan have been continually attempting to diplomatically engage with Ethiopia in discussions regarding the allocation of Nile water. After the announcement of the GERD’s construction in 2011, the three states came together to sign the 2015 Declaration of Principles (DoP), which grants Ethiopia’s right to develop the dam over the Blue Nile, but did not provide any specific binding regulations and terms on how the reservoirs of the GERD would be operated, filled, and released.

However, it was only later in 2019 when Egypt realised the DoP was insufficient in guaranteeing its water security and began facilitating more talks with Ethiopia under third-party mediation. There has first been talks moderated by the United States and the World Bank, that drew attention to the use of the reservoirs and referred to existing regional frameworks such as in the Senegal River basin, but the draft was incomplete with Ethiopian withdrawal last minute; arguing that the treaty severely constrained its sovereign right to develop the Blue Nile water resources and insisted on non-binding guidelines instead.

With western and international arbitration failing, the African Union (AU) became the principal medium for negotiation between both sides, with Ethiopia emphasizing “African solutions to African problems”, while Egypt declaring the negotiation to be deadlocked and impasse; diplomatic relations between Egypt and Ethiopia remain tense, with Egyptian authorities asserting that Ethiopia is neglecting international law and the operation of the GERD poses a threat to downstream water security. The contention surrounding the GERD remains as one of the significant hydro-political issues to this day.

US – Mexico Water Dispute

On December 8, 2025, President Donald Trump issued a warning to Mexican authorities, stating that they must release 200,000 acre-feet of water by December 31 or face a 5% tariff imposed by the US government to enforce compliance.

The dispute between the United States and Mexico originated from the 1944 water treaty that was signed between the two signatories; which was a document in which Mexico would be given 1.5 million acre-feet of water from the Colorado River annually, while the United States will receive most of the Rio Grande, with Mexico obligated to also deliver 350,000 additional acre-feet of water from the Rio Grande's tributaries every year over a five year cycle.

However, during the 2020-2025 delivery cycle, it is reported that Mexico has fallen behind in providing 865,000 acre-feet of water, which is nearly half of the expected volume. President Trump claims that the Mexican authorities have been "unfair" to US farmers, particularly in Texas and New Mexico, and demands that Mexico "has an obligation to fix this now". Meanwhile, the Mexican government had responded by attributing its failure to reasons such as climate constraints and the fact that the northern parts of Mexico have been suffering severe droughts, and asserted that the volume of water to be delivered must respect the availability of the resource in the first place.

This led to the withholding of US water deliveries to Mexico from the Colorado River to Tijuana for the first time in 80 years, a delivery that is not treaty-mandated but rather a voluntary commitment by the US to supply Tijuana during periods of drought, exacerbating tensions between the two parties. Mexico responded to the denial of delivery with measures to offset the shortfall, including plans for desalination, contingency measures, and improvements to aqueducts.

The result was institutional intervention from the International Boundary and Water Commission, allowing both sides to reach a peaceful conclusion without further escalation of trade tensions. Mexico committed to accelerating its water release, while the United States accepted phased compliance instead of immediate repayment.

Major Parties Involved

Ethiopia

Ethiopia is one of the key upstream riparian nations that controls the Nile River in East Africa. Although the Nile River is long, it does not transport much water proportionate to its length. Many nations in Africa rely on the Nile for growth and development, and it now supports hundreds of millions of individuals who reside in these nations. The Nile River has two main tributaries, the Blue Nile and the White Nile; the Blue Nile originates in Ethiopia, giving the nation significant influence over the downstream flow. Currently, Ethiopia has not used up much of the Nile's water, as conflict and economic restrictions in the 20th century have prevented many upstream nations, such as Ethiopia, from developing their resources; however, in the future, Ethiopia's projects appear growingly ambitious, and they could seriously impact downstream water resources. (Climate Diplomacy) For example, in 2011, Ethiopia announced the construction of a 6,000 MW hydroelectric dam on the Blue Nile, a tributary within its territory, which led to heightened tensions between Ethiopia, Egypt, and Sudan. Ethiopia, therefore, has a critical role in preventing conflict as a major Upstream Riparian Nation. (Climate Diplomacy)

China

China is another key upstream riparian nation, and is where the Mekong River originates. From China, the river flows through several other Asian nations, including Myanmar, Laos and Cambodia. The Mekong River Commission was established in 1995; however, they struggle to make progress as China has never been a full member. (Mekong River Commission (MRC)) In recent years, China has initiated plans to construct dams on the Mekong to harness hydropower, and the Country has stated that the negative effects on downstream nations are negligible.

Furthermore, China states that the effects of these dams may even be beneficial for flood control. Despite this, many downstream nations have raised concerns about water shortages and habitat destruction. As the years pass, this issue may intensify, as erratic weather patterns caused by climate change are likely to exacerbate these problems. So far, China shows no signs of stopping dam construction, and it firmly believes that its state sovereignty allows it to construct dams within its own territories. Unfortunately, this is likely to raise tensions between China and downstream nations, which may become more severe in the future. Thus, China plays a significant role in this issue as a major Upstream Riparian Nation. (Climate Diplomacy)

India

India began constructing the Kishanganga hydroelectric power plant (KHEP) in 2007, with the goal of diverting water into a generator house located in Kashmir, a long-contested area between India and Pakistan. This was India's attempt to redirect water into its territory before the river re-entered Pakistan, and this redirection went against the Indus Water Treaty. Eventually, Pakistan sought legal action in 2013 at the Permanent Court of Arbitration in The Hague, where it was decided that India could construct this plant as long as the downstream flow of water to Pakistan was not affected. This allowed India to harness the power of the river with lessened impacts on Pakistan. Due to this conflict regarding the Indus waters, India is a key upstream riparian nation which wishes to benefit from the resources of this river. (Climate Diplomacy, "Conflicts Surrounding the Kishanganga Dam")

Egypt

Egypt has been harnessing the Nile's downstream flow for thousands of years, and its usage of the Nile has only recently been heavily contested. Throughout history, Egypt's arid climate and desert-dominated geography have resulted in limited rainfall, leading to its reliance on the Nile as a primary water source. Thus, upstream infrastructural developments pose a significant risk to Egypt, and historically, Egypt has been strongly against any project that would decrease downstream water flow. In recent years, political unrest and other factors have led to Egypt losing its regional influence, meaning it can no longer easily influence upstream states. (Climate Diplomacy) As upstream states grow economically and politically, Egypt

is likely to be negatively impacted, posing a risk to its agricultural industry. Therefore, Egypt is a major stakeholder in this issue and seeks to protect the rights of downstream riparian nations.

Cambodia

As a downstream riparian nation heavily reliant on the Mekong River, Cambodia is concerned about construction projects upstream and their potential impact on its stretch of the river. Cambodia and other downstream nations have already been affected by dams or similar projects upstream, facing negative effects such as agricultural devastation and flooding. One area of particular concern to Cambodia is the Tonle Sap, a system of rivers and lakes within Cambodia that provides 2/3s of the inland fish catch. As more dams are constructed upstream, the food security of these areas within Cambodia will be negatively impacted; combined with climate change, these pose significant dangers to Cambodia's future livelihood. Cambodia wants upstream nations, such as China, to increase their transparency and consider the long-term consequences, making Cambodia a key stakeholder in this issue. (Climate Diplomacy)

Pakistan

In 2007, India began construction on the KHEP on the Kishanganga River, prompting Pakistan to express concern about the potential impact on water availability downstream. The dam would have redirected river flow to India through the contested region of Kashmir, greatly concerning Pakistan, as it believed this would go against the Indus Water Treaty. Pakistan attempted to resolve this conflict by raising it at the Permanent Court of Arbitration, where it was ruled that India could continue constructing the KHEP if it did not negatively impact Pakistan. Previously, Pakistan strongly opposed the KHEP, as it would have impacted their agricultural, electrical, and economic objectives. As a downstream riparian nation threatened by potential construction projects, Pakistan is a key stakeholder in this issue. (Climate Diplomacy, "Conflicts Surrounding the Kishanganga Dam")

UN Water

UN Water is a coordination mechanism of UN entities and nations that focuses on water-related issues. Established in 2003, it is now crucial to organise international cooperation in many areas of the water issue. UN Water tracks many water-related problems, including Indicator 6.5.2, "Proportion of transboundary basin area with an operational arrangement for water cooperation." By establishing frameworks for international cooperation, UN Water clarifies criteria for meaningful arrangements while also linking to other UN bodies that may be relevant. This makes UN Water important to this issue, as the organisations it contains coordinate and facilitate discussions and solutions.

Timeline of Events

Date	Description of Event
1960	Indus Water Treaty
1967 ~ 1973	Sahel Drought
1973	Helmand River Treaty
1992	UNCE Water Convention
1994	Israel-Jordan Peace Treaty
1995	Mekong Agreement
1997	UN Watercourses Convention
1998 ~ 2001	Horn of Africa Drought
1999	Nile Basin Initiative
2000	Catastrophic Mozambique Flooding
2006 ~ 2010	Syrian Drought
2011	South Sudanese Independence
2011	Arab Spring Uprisings
2018	Cape Town "Day Zero" Crisis
2019	Pulwama Attack
2020	East African Locust Crisis
2021	Iran – Afghanistan Border Skirmishes
2025	India – Pakistan Tensions

Previous Attempts to Resolve the Issue

Indus Water Treaty

A treaty was signed between India and Pakistan in 1960 regarding the hydrological use of the Indus basin, in which three of the Eastern Rivers, including Ravi, Beas, and Sutlej were designated for India's unrestricted use, whilst three of the Western Rivers, such as Indus, Jhelum, and Chenab being allocated towards Pakistan predominantly, with a minority of water being used for India under non-consumptive purposes or for limited agricultural and hydroelectric uses with strict constraints.

Included in the treaty was the establishment of a Permanent Indus Commission, which later became not only a bilateral channel of communication between the two parties for resolving disputes arising from the Indus basin, but also served as an inspection body for providing technical support in the hydrological management of the region. When discourse occurs between two sides, the system moderates and aims to facilitate peace through three different levels: the Commission, followed by a Neutral Expert, and finally a Court of Arbitration, in which an arbitrator is appointed to adjudicate the situation.

The Indus Water Treaty has faced many threats across the decades since it's designation; with massive flooding due to the acceleration of glacier sheet melting in 2010, and political escalation originating from the 1965 Indo – Pakistan War, the 1971 Bangladesh Liberation War, the 1999 Kargil Conflict, the 2001 Twin Peaks Crisis, the

2008 Mumbai Attacks, the 2016 Uri Attack and India's political proclamation of "blood and water cannot flow together", the 2019 Pulwama Attack, and lastly the recent border tensions and skirmishes during 2025. Despite numerous threats to its stability, the Indus Water Treaty remains intact; yet, it continues to be a subject of intense contention between the two parties.

Figure 3, Indian soldiers on a search operation on Wednesday after the aftermath of the Kashmir Attack

Recent years of strategic weaponisation of upstream infrastructure, together with rising populations, increased landmass required for irrigation and climate change altering the river flow, have led to political tensions, further amplifying tensions between India and Pakistan. A notable example of how the development of upstream infrastructure has stepped up discord between riparian's includes India's hydroelectric plans at Baglihar, Kishanganga, Ratle, Sawalkot, and Pakal Dul, which, despite being legal within the treaty, still receive criticisms from Pakistan regarding the architectural designs of the dams, such as spillway height, drawdown capabilities, and storage volume of reservoirs, et cetera, giving India opportunities to control flow time for irrigation. This consequently creates frequent diplomatic collapses between the two parties, such as Pakistani distrust towards India, and India interpreting such actions as hostility and an attempt to stall their infrastructural modernisation. Additionally, and more alarmingly, both parties have consistently over the span of decades shifted battalions and military forces into the contested regions of Kashmir, which has furthermore exacerbated the dispute.

Nile Basin Initiative

The Nile Basin Initiative (NBI) was a collective effort and movement facilitated by Nile basin nations such as Burundi, the Democratic Republic of Congo, Egypt, Ethiopia, Kenya, Rwanda, South Sudan, Sudan, Tanzania, and Uganda, alongside Eritrea participating as an observer state, to promote greater socio-economic development within the region while also preserving regional peace by equitable utilization of the Nile river.

Figure 4, map of water scarcity in the Nile basin region, from Research Gate

The current established and ongoing vision of the NBI is a 10-year plan between 2017 and 2027, with an emphasis on key objectives such as water, food, and energy security, environmental sustainability, climate adaptation and resilience, and enhanced governance and dialogue over transboundary water resources. In practice, this would mean expanding and improving current irrigation systems, attracting investments for infrastructure projects, enhancing water use efficiency, increasing regional power trade through hydroelectricity, and protecting local biodiversity and ecosystems while maintaining water quality.

The NBI is divided into three divisions and programs, each responsible for different purposes and tailored to different member states. For instance, the Shared Vision Program (SVP) strives to build institutions focused on hydrology, providing training as well as shared data and intelligence on the management of the Nile. Furthermore, it partakes in stakeholder engagement and funnels investments towards the cause. Meanwhile, there's the Eastern Nile Subsidiary Action Programme (ENSAP), which is targeted at nations such as Egypt, Ethiopia, Sudan, and South Sudan, primarily involved in efforts such as joint planning for irrigation, flood management, and hydropower in the Eastern Nile sub-basin. Lastly, the Nile Equatorial Lakes Subsidiary Action Programme (NELSAP) covers the remaining southern equatorial member states, assisting them with the construction of water infrastructure, equitable utilisation of the natural resources, and the facilitation of trade, within the great lakes that are upstream of the Nile.

Overall, this initiative remains crucial to this day in the Horn of Africa and Northeastern Africa, promising the fundamental solidarity of peace within the region by collaborative management in the Nile water resources; despite the residual of tensions between upstream and downstream nations along the Nile, such as Egypt and Sudan entering disputes with other upstream nations that have more abundant water resources around the legal principles of the NIB. Regardless, this collective effort remains one of the prime examples of how nations have succeeded in working towards the greater governance of transboundary water resources.

Mekong Agreement

The Mekong Agreement was signed between member states, including Cambodia, Laos, Thailand, and Vietnam, as well as upstream nations such as China and Myanmar, which serve as “dialogue partners” for the Mekong River Commission. This agreement serves as the legal foundation for the sustainable development of the Mekong basin among these Southeast Asian countries.

The agreement primarily aims to enhance mutual benefits within the region, particularly in sectors such as hydropower, fisheries, navigation, irrigation, tourism, and flood management. Member nations are required to notify and consult other countries prior to the construction of hydroelectric projects that may affect water usage and equitable utilisation across other riparian states.

The MRC is composed of three bodies and branches of organization; the Council, which represents the various water and environment ministers from member states, in which ministers work together in approving high level decisions, documents, and joint projects regarding water use and any disputes that may arise from it; the MRC Joint Committee, which is the department responsible for the formulation of various plans and policies, to then be presented to the Council for final approval, and also the management and supervision of such plans to be carried out; and lastly, the MRC Secretariat, executes all implemented procedures, monitors hydroelectrical plants, and conducts assessments on sectors such as fishing and hydropower if they meet the regulatory requirements.

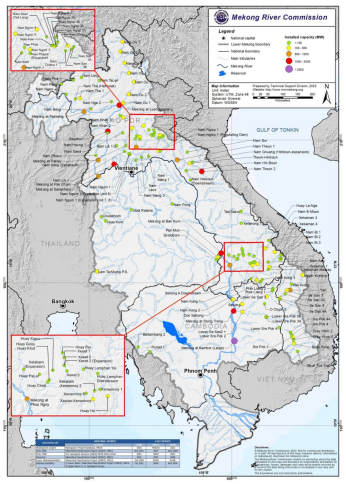

Figure 5, map of hydropower facilities in the Mekong basin, from the MRC

However, the Mekong Agreement is not flawless. Vague regulations and non-binding properties make enforcement of compliance less effective, alongside the exclusion of China as a member state, leading to manipulated flow time and ecological damage downstream. Additionally, aggressive dam building and hydropower competition between member states, such as Laos prioritising revenue by exporting power to Thailand over mutual benefits in the basin, while downstream members, including Thailand and Cambodia, experience declines in fisheries due to artificial dry seasons and floods.

Israel-Jordan Peace Treaty

After Arab invasions since Israel’s independence, it was only until October 26th, 1994, that both nations ended their state of war with each other and were able to make a conclusion to their 46 years of formal belligerency.

The treaty designates specific volumes of consumption with the Jordan and Yarmouk Rivers that flows between Israel and Jordan for both parties, including both groundwater and surface water, with Israel pumping 12 million cubic meters (MCM) of water during the summer from May 15 ~ October 15, with Jordan having free and unrestricted access to the rest; while in the winter from October 16 ~ May 14, it works vice versa with Jordan conceding water flow to Israel whilst only consuming 12 MCM.

Legislative mechanisms include the Joint Water Committee (JWC), in which three members represent both sides, and infrastructure is monitored and supervised, such as the Yarmouk diversion dam. Israel oversees its commitment and operations in delivering an additional 50 MCM of potable water towards Jordan, while Jordan manages water wells and pumps for Israel.

The terms of the peace treaty have ensured consistent, stable and predictable transfers of Israeli water to Jordan, and assisted in the stabilization of the Middle Eastern region, and dialogue channels such as the JWC enables effective communication and negotiation between both parties; particularly in aspects such as adjusting releases and flow time during dry seasons and droughts, and making emergency increases in deliveries for water possible.

However, experts view the peace treaty as more favourable towards Israel than Jordan. Debated facets include Jordan's financial and technical restrictions on greater storage reservoirs for the Yarmouk through stricter quality standards. Under the treaty in Annexe II, which outlines the utilisation of the waters within the Jordan basin, some plans are still not fully operationalised and implemented, including full development of Jordanian water storage. Additionally, recent geopolitical instability within the Middle East, including the Israel-Palestine conflict, has received domestic backlash in Jordan, leading to JWC meetings being delayed and new projects being put on pause.

Furthermore, this treaty is bilateral and does not address or mention any collaboration with other riparian states, such as Palestine, Lebanon, and Syria, and only remains as a successful bilateral initiative in resolving disputes in the governance of transboundary resources.

Possible Solutions

Encouraging transparency and joint management

One of the most effective aspects of resolving issues regarding transboundary water resources is encouraging transparency among all stakeholders, namely, the nations directly affected by the issue. This can be achieved by creating databases and conducting scientific research to quantify the environmental impacts of all nations and their effects on other stakeholders. It is essential that nations do not withhold crucial information regarding the environmental impact on their water resources, as this could have detrimental long-term consequences when climate change becomes more severe. Furthermore, this data from these transparency measures should be redirected to projects which are jointly managed by all riparians. This could include hydropower projects or even new agricultural technologies that efficiently use the water resource. (Sarfaraz et al.)

Creating and implementing new technology

Another effective solution that would lessen tension is implementing new technology that would reduce the need for such large amounts of water. This could include investing in other energy harvesting

methods that do not include water, such as solar power and wind energy. (Sustainability Directory) Moreover, nations could cooperate in developing technologies that help mitigate the negative impact of construction projects, which in turn could assist in preventing floods or other detrimental events. This would alleviate the concerns of downstream riparians, as their economy wouldn't be compromised as much, and would also foster peace with upstream riparians.

Strengthening existing legal frameworks

Strengthening existing legal frameworks is a method that could be successful in combating issues related to political tension caused by transboundary waters; however, it is essential to note that many of these methods may have been employed before and may not be effective. Thus, it is important to realise that changes to these frameworks may be required for them to be effective. There have been several treaties and legal documents drafted and passed already that pertain to this issue (such as the 1997 UN Watercourses Convention), so delegates must be wary to improve on past attempts and not merely just repeat them. It would be helpful to come to compromises between nations and ensure that most nations join the treaty, as that has been a pitfall of many previous attempts. (UN, 2011)

Creating tailored legal frameworks

For specific situations or conflicts over water resources between nations, establishing new legal frameworks may be beneficial. This would be effective because, in the long term, it would prevent conflict. A problem with many treaties is that they do not target specific situations, leading to excessive treaty violations. Thus, creating an effective mechanism for drafting region-specific treaties would be important for reducing tensions. It would be important to include neutral facilitators, other third-party organisations, and the governments of all affected nations when these treaties are created. Upstream riparians and downstream riparians would both have to agree to certain conditions that protect the water resources and ensure all nations can access them. This would alleviate the tension and fear that mount when water resources are compromised.

Bibliography

Climate Diplomacy. "Conflicts Surrounding the Kishanganga Dam." *Climate-Diplomacy*, climate-diplomacy.org/case-studies/conflicts-surrounding-kishanganga-dam#fact-sheet-toc-conflict-history. Accessed 7 Dec. 2025.

Climate Diplomacy. “Dam Projects and Disputes in the Mekong River Basin.” *Climate-Diplomacy.org*, 2024, climate-diplomacy.org/case-studies/dam-projects-and-disputes-mekong-river-basin.

Climate Diplomacy. “Dispute over Water in the Nile Basin.” *Climate-Diplomacy.org*, climate-diplomacy.org/case-studies/dispute-over-water-nile-basin.

Climate Diplomacy. “Disputes over the Grand Ethiopian Renaissance Dam (GERD).” *Climate-Diplomacy.org*, climate-diplomacy.org/case-studies/disputes-over-grand-ethiopian-renaissance-dam-gerd.

Concern USA. “The Global Water Crisis, Explained.” *Concernusa.org*, 2024, concernusa.org/news/global-water-crisis-explained/.

“Fact Sheet: The Indus Waters Treaty 1960 and the Role of the World Bank.” *World Bank*, www.worldbank.org/en/region/sar/brief/fact-sheet-the-indus-waters-treaty-1960-and-the-world-bank. Accessed 6 Dec. 2025.

“History.” *Nile Basin Initiative*, nilebasin.org/about-us/history#:~:text=On%2022nd%20February%201999%2C%20in,the%20new%20transitional%20institutional%20arrangement. Accessed 6 Dec. 2025.

Kasimbazi, Emmanuel. *Overview of Key Principles of Equitable and Reasonable Use, and No Significant Harm*. UNECE, 2025, unece.org/sites/default/files/2025-01/Overview%20of%20key%20principles%20of%20equitable%20and%20reasonable%20use%2C%20and%20no%20significant%20harm.pdf.

Magazine, Diplomat. “Legal Implications of India’s Notices to Pakistan on Indus Water Treaty.” *Diplomat Magazine*, 15 Dec. 2024, diplomatmagazine.eu/2024/12/15/legal-implications-of-indias-notices-to-pakistan-on-indus-water-treaty/.

Mekong River Commission (MRC). “Mekong River Commission (MRC).” *MRC Mekong*, 2024, www.mrcmekong.org/.

Mekong River Commission (MRC). *Understanding the 1995 Mekong Agreement and the Five MRC Procedures: A Handbook*. MRC Secretariat, May 2020, www.mrcmekong.org/wp-content/uploads/2020/05/Understanding-the1995-Mekong-Agreement-and-the-Five-MRC-Procedures-A-Handbook.pdf.

“Our Goals.” *Nile Basin Initiative*, nilebasin.org/our-goals. Accessed 7 Dec. 2025.

Pech, Sokhem. *Mekong Agreement Now and Then: How the 1995 Agreement Was Developed, What Its Provisions Have Covered Since Then, and Where Gaps Exist*. UNECE, 7 Nov. 2023, unece.org/sites/default/files/2023-11/5.%20Sokhem%20Pech_Mekong%20Agreement%20now%20and%20then.pdf.

Sarfaraz, M. Umer, et al. “Data Sharing in Transboundary Water Management.” *Frontiers in Water*, vol. 4, 12 Sept. 2022, doi.org/10.3389/frwa.2022.982605.

"Stronger Together: 20 Years of Cooperation Around the Nile." *World Bank*, www.worldbank.org/en/news/feature/2019/02/22/stronger-together-20-years-of-cooperation-around-the-nile. Accessed 7 Dec. 2025.

Sustainability Directory. "Which Energy Sources Consume the Least Water? → Question." *Energy → Sustainability Directory*, Dec. 2025, energy.sustainability-directory.com/question/which-energy-sources-consume-the-least-water/. Accessed 7 Dec. 2025.

Sustainable Socio-Economic Development through the Equitable Utilisation of, and Benefit from, the Common Nile Basin Water Resources. United Nations Department of Economic and Social Affairs, sdgs.un.org/partnerships/sustainable-socio-economic-development-through-equitable-utilization-and-benefit. Accessed 7 Dec. 2025.

Treaty between the United States of America and Mexico Relating to the Utilisation of the Waters of the Colorado and Tijuana Rivers and of the Rio Grande. 1944, gis.nacse.org/tfdd/tfdddocs/archiveApril2010/55ENG.htm. Accessed 26 Dec. 2025.

Treaty of Peace Between the State of Israel and the Hashemite Kingdom of Jordan. Peace Treaty, 26 Oct. 1994. United Nations Peacemaker, peacemaker.un.org/sites/default/files/document/files/2024/05/il20jo941026peacetreatyisraeljordan.pdf.

"Trump Threatens Mexico with 5% Tariff Over Water Treaty." *Ground News*, 8 Dec. 2025, ground.news/article/trump-threatens-mexico-with-5-tariff-over-water-treaty_62b27e.

United Nations. *Convention on the Law of the Non-navigational Uses of International Watercourses*. UNGA, 21 May 1997, legal.un.org/ilc/texts/instruments/english/conventions/8_3_1997.pdf.

United Nations. "What Is Water Security?" *UN-Water*, 8 May 2013, www.unwater.org/publications/what-water-security-infographic.

United Nations Environment Programme. "Integrated Water Resources Management." *UNEP*, 16 Nov. 2023, www.unep.org/topics/fresh-water/water-resources-management/integrated-water-resources-management.

World Bank. *Mekong River Basin: Challenges and Opportunities for Sustainable Development*. World Bank, 2007, documents1.worldbank.org/curated/en/277291468224093302/pdf/411660EAP0Meko1ase1study1201PUBLIC1.pdf.

World Bank. "World Annual Fresh Water Losses Could Supply 280 Million People." *World Bank*, 4 Nov. 2025, www.worldbank.org/en/news/press-release/2025/11/04/world-annual-fresh-water-losses-could-supply-280-million-people. Accessed 9 Dec. 2025.

Wallis, William, and Heba Saleh. "Africa's Largest Dam Triggers Alarm down the Nile." *Financial Times*, 8 Sept. 2025.

Zahran, Motaz. “Only Washington Can Save the Renaissance Dam Negotiations Now.” Foreign Policy, 1 Jan. 2026, https://foreignpolicy.com/2021/04/29/gerd-renaissance-dam-negotiations-biden-ethiopia-egypt/.

Security Council Presidential Statement Encourages Egypt, Ethiopia, Sudan to Resume Talks on Grand Renaissance Dam Project | UN Meetings Coverage and Press Releases. https://press.un.org/en/2021/sc14637.doc.htm?utm_source=chatgpt.com. Accessed 30 Dec. 2025.

“Trump Threatens Mexico with 5% Tariff Over Water Treaty.” Ground News, 8 Dec. 2025, https://ground.news/article/trump-threatens-mexico-with-5-tariff-over-water-treaty_62b27e.